Completing the Square - Mark scheme

Algebra · Edexcel IGCSE Mathematics A (4MA1) · Year 10

Question 1 *Practice question written for this worksheet (not from a past paper)*

Write $x^2 + 4x + 7$ in the form $(x + a)^2 + b$.

Half of 4 is 2: $(x + 2)^2 - 4 + 7$

Answer: $(x + 2)^2 + 3$

Question 2 *Practice question written for this worksheet (not from a past paper)*

Write $x^2 - 10x + 40$ in the form $(x + a)^2 + b$.

Half of -10 is -5: $(x - 5)^2 - 25 + 40$

Answer: $(x - 5)^2 + 15$

Question 3 *Practice question written for this worksheet (not from a past paper)*

Write $x^2 + 8x - 3$ in the form $(x + a)^2 + b$.

$(x + 4)^2 - 16 - 3$

Answer: $(x + 4)^2 - 19$

Question 4 *Practice question written for this worksheet (not from a past paper)*

Write $x^2 - 5x + 1$ in the form $(x + a)^2 + b$.

$\left(x - \frac{5}{2}\right)^2 - \frac{25}{4} + 1$

Answer: $(x - 5/2)^2 - 21/4$

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Question 5 *Practice question written for this worksheet (not from a past paper)*

Solve $x^2 + 6x - 2 = 0$ by completing the square.

Give your answers in the form $a \pm \sqrt{b}$.

$(x + 3)^2 - 9 - 2 = 0$, so $(x + 3)^2 = 11$

$x + 3 = \pm\sqrt{11}$

Answer: x = -3 ± √11

Question 6 *Practice question written for this worksheet (not from a past paper)*

Write $3x^2 + 12x + 5$ in the form $a(x + p)^2 + q$.

$3(x^2 + 4x) + 5 = 3[(x + 2)^2 - 4] + 5$

$3(x + 2)^2 - 12 + 5$

Answer: $3(x + 2)^2 - 7$

Question 7 *Practice question written for this worksheet (not from a past paper)*

Write $x^2 - 6x + 11$ in the form $(x + a)^2 + b$.

Hence write down the minimum value of $x^2 - 6x + 11$ and the value of x where it

$(x - 3)^2 - 9 + 11 = (x - 3)^2 + 2$

A square is never negative, so the minimum is 2, when $x = 3$

Answer: minimum 2 at x = 3

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Completing the Square - Mark scheme

Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2023 · Question 19

19	$3(x^2-2x).....$ or $3(x^2-2x+.....)$ oe or		3	M1	(where …… is any number or no number)
	$3(x-1)^2.....$ or $3[(x-1)^2.....]$ oe			M1	(where …… is any number or no number)
	Correct answer scores full marks (unless from obvious incorrect working)	$3(x-1)^2+2$		A1	(if student continues to solve a quadratic equation, ISW)
					Total 3 marks
Alternative mark scheme for 19					
19	$ax^2-2abx+ab^2+c$		3	M1	for multiplying out $a(x-b)^2+c$ to obtain $ax^2-2abx+ab^2+c$ oe
	Any 2 of the following: $a=3$ or $2ab=6$ or $ab^2+c=5$ oe			M1	for equating coefficients with any 2 of $a=3$ or $2ab=6$ or $ab^2+c=5$ oe seen
	Correct answer scores full marks (unless from obvious incorrect working)	$3(x-1)^2+2$		A1	(if student continues to solve a quadratic equation, ISW)
					Total 3 marks

Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 21

21	$5(x-2)^2.....$ **or** $5[(x-2)^2.....]$ **or** $5\left(x+\frac{-20}{5\times2}\right)^2+....$ **or** $5(x-b)^2+c$		3	M1	for a start to completing the square **or** correct substitution into $a\left(x+\frac{b}{2a}\right)^2+....$ from the formula $a\left(x+\frac{b}{2a}\right)^2-\frac{(b)^2}{4a}+c$ **or** $a=5$ embedded in an incorrect final answer in the form $5(x-d)^2+e$ (must be these signs)
	$5[(x-2)^2-2^2].....$ **or** $5[(x-2)^2-4].....$ **or** $5[(x-2)^2-2^2.....]$ **or** $5[(x-2)^2-4.....]$ **or** $5(x-2)^2-20.....$			M1	for correctly completing the square but terms do not need to be simplified and 23 may or may not be present correct simplification **or** of the first two parts of $a\left(x+\frac{b}{2a}\right)^2-\frac{(b)^2}{4a}(+c)$ **NB: Please refer to ALT mark scheme for comparison of coefficients method**
	Correct answer scores full marks (unless from obvious incorrect working)	$5(x-2)^2+3$		A1	oe eg $3+5(x-2)^2$ (if student continues to solve a quadratic equation, ISW)
					Total 3 marks

21 ALT	$ax^2-2abx+ab^2+c$		3	M1	for multiplying out $a(x-b)^2+c$ **or** $a=5$ embedded in an incorrect final answer in the form $5(x-d)^2+e$ (must be these signs)
	$-2ab=-20$ **or** $2ab=20$ **or** $ab^2+c=23$			M1	for equating coefficients
	Correct answer scores full marks (unless from obvious incorrect working)	$5(x-2)^2+3$		A1	oe eg $3+5(x-2)^2$ (if student continues to solve a quadratic equation, ISW)
					Total 3 marks

Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2019 · Question 20

Question		Working	Answer	Mark	Notes
20	(a)	$3(x^2-4x)+7$ or $3\left(x^2-4x+\frac{7}{3}\right)$ $3((x-2)^2-4)+7$ or $3\left((x-2)^2-4+\frac{7}{3}\right)$ or $3(x-2)^2-12+7$	$3(x-2)^2-5$	3	M1 or expanding $a(x^2+2bx+b^2)+c$ M1 $-12=2ab$ or $7=ab^2+c$ A1 or $a=3, b=-2, c=-5$
	(b)		$x=2$	1	B1 ft from (a)

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2024 · Question 20

20	(a)	$2\left(x^2-\frac{11}{2}x\right)+\ldots$ **or** $2\left(x^2-\frac{11}{2}x+\ldots\right)$ oe		3	M1 for taking out a factor of 2
		$2\left[\left(x-\frac{11}{4}\right)^2-\frac{11^2}{4^2}\right]+\ldots$ **or** $2\left[\left(x-\frac{11}{4}\right)^2-\frac{11^2}{4^2}+\ldots\right]$ oe			M1 for correctly completing square
		Correct answer scores full marks (unless from obvious incorrect working)	$2\left(x-\frac{11}{4}\right)^2-\frac{49}{8}$		A1 oe, eg $2(x-2.75)^2-6.125$ allow $a=2$, $b=\frac{11}{4}$ oe, $c=\frac{49}{8}$ oe if no other marks awarded, award SCB1 for $2\left(x-\frac{11}{4}\right)^2+\ldots$
		ALTERNATIVE			
ALT	(a)	$ax^2-2bax+b^2a-c$		3	M1 for correctly expanding $a(x-b)^2-c$ to give $ax^2-2bax+b^2a-c$
		$-2ba=-11$ or $2ba=11$ and $b^2a-c=9$			M1 for setting up 2 equations using the coefficient of x and the numerical term
		Correct answer scores full marks (unless from obvious incorrect working)	$2\left(x-\frac{11}{4}\right)^2-\frac{49}{8}$		A1 oe, eg $2(x-2.75)^2-6.125$ allow $a=2$, $b=\frac{11}{4}$ oe, $c=\frac{49}{8}$ oe if no other marks awarded, award SCB1 for $2\left(x-\frac{11}{4}\right)^2+\ldots$
	(b)		$\left(\frac{23}{4},-\frac{49}{8}\right)$	2	B2ft oe, eg (5.75, −6.125) (B1ft for one correct coordinate)
					Total 5 marks

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2022 · Question 23

23	(a)	$2(x^2-6x)+3$ **or** $2(x^2-6x+\frac{3}{2})$		3	M1 **or** for one of a, b or c correct **OR** expanding $a(x^2+2bx+b^2)+c$
		$2\left[(x-3)^2-9\right]+3$ **or** $2\left[(x-3)^2-3^2+\frac{3}{2}\right]$ oe			M1 **or** for two of a, b or c correct **OR** $-12=2ab$ **or** $3=ab^2+c$
			$2(x-3)^2-15$		A1 accept $a=2$, $b=-3$, $c=-15$
	(b)		$(-1,-15)$	2	B2ft eg accept [their $-b-4$] for the x-coordinate or [their c] for the y-coordinate (B1 ft for one correct coordinate)
					Total 5 marks

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2026 · Question 24

Question		Working	Answer	Mark	Notes
24	(a)	$-3(x+4)^2$ ..... **or** $-3\left[(x+4)^2 .....\right]$ **or** $-3\left(x+\frac{-24}{2\times-3}\right)^2+....$		3	M1 for a start to completing the square **or** correct substitution into $a\left(x+\frac{b}{2a}\right)^2+....$ from the formula $a\left(x+\frac{b}{2a}\right)^2-\frac{(b)^2}{4a}+c$
		$-3\left[(x+4)^2-4^2\right]$ ..... **or** $-3\left[(x+4)^2-4^2 .....\right]$ **or** $-3\left[(x+4)^2-16 .....\right]$ **or** $-3\left(x+\frac{-24}{2\times-3}\right)^2-\frac{(-24)^2}{4\times-3}...$ **or** $-3(x+4)^2+48.....$			M1 for correctly completing the square but terms do not need to be simplified and 10 may or may not be present **or** correct simplification of the first two parts of $a\left(x+\frac{b}{2a}\right)^2-\frac{(b)^2}{4a}(+c)$
		Correct answer scores full marks (unless from obvious incorrect working)	$58-3(x+4)^2$		A1 oe eg $-3(x+4)^2+58$
	(b)		$(-3, 58)$	1	B1 ft their expression in part (a)
	(c)		$x=-3$	1	B1 ft their expression in part (a) **or** ft their x coordinate in part (b)
					Total 5 marks

Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · November 2020 · Question 24

24	(a)	$-3(x^2-4x)+7$ or $-3\left(x^2-4x-\frac{7}{3}\right)$		4	M1	for factorising the expression to find b or $b=-3$ stated or shown clearly in answer.
		$-3\left[(x-2)^2 .....\right]$ or $c=-2$ $-3\left[(x-2)^2-4\right]+7$ or $-3\left[(x-2)^2-4-\frac{7}{3}\right]$			M1	or for c shown clearly in answer.
		$-3(x-2)^2+12+7$ or $-3\left[(x-2)^2-\frac{19}{3}\right]$			M1	fully correct method.
			$19-3(x-2)^2$		A1	for $19-3(x-2)^2$ oe
	(b)		$(2, 19)$	1	B1 ft	dep on M1 in part (a) answer must follow answer from (a) if given
						Total 5 marks

Alternative mark scheme for 24						
24	(a)	$a+bx^2+2bcx+bc^2$		4	M1	for multiplying out $a+b(x+c)^2$ to obtain $a+bx^2+2bcx+bc^2$ oe
		$b=-3$ or $2bc=12$ or $a+bc^2=7$ oe			M1	for equating coefficients
		$b=-3$ and $c=-2$			M1	for correctly finding b and c
		$a=19$	$19-3(x-2)^2$		A1	for $19-3(x-2)^2$ oe
	(b)		$(2, 19)$	1	B1 ft	dep on M1 in part (a)
						Total 5 marks

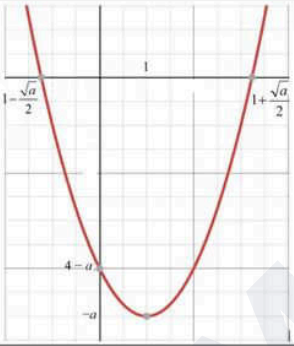
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CHALLENGE QUESTIONS

Question 15 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2018 · Question 24*

24	(a)		$11-(x+2)^2$	2	M1	For $11-(x+q)^2$ or $p-(x+2)^2$
					A1	fully correct, accept $p=11$, $q=2$
	(b)	$(y+3+2)^2=11$ or $11-(y+3+2)^2$		3	M1	substituting $x=y+3$ into their $p-(x+q)^2$
		$y+3+2=\pm\sqrt{11}$			M1	
			$-5\pm\sqrt{11}$		A1	Both answers correct, ft their answer from (a) eg $-(3+"q")\pm\sqrt{"p"}$
	ALT (b)	**Alternative scheme**			M2	for $-y^2-10y-14=0$ or $y^2+10y+14=0$
			$-5\pm\sqrt{11}$		A1	cao, both values correct
	(c)		$(-1, 3)$	1	B1	cao
						Total 6 marks

Question 16 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2020 · Question 20*

20	graph drawn in shape of a quadratic with a minimum in any quadrant		4	M1 for a quadratic with a minimum
	$x=1$, $y=4(1-1)^2-a$			M1 for finding the turning point (may be seen marked on the graph as $(1, -a)$)
	$x=1\pm\sqrt{\frac{a}{4}}$ oe **or** $y=4-a$			M1 for finding one of the intercepts (or award for any one correct coordinate shown on graph) $(0, 4-a)$ or $(1+\frac{\sqrt{a}}{2}, 0)$ or $(1-\frac{\sqrt{a}}{2}, 0)$ Note: The 0's can be ignored (as shown in the diagram)
		Correct graph		A1 for a fully correct graph • quadratic shape with minimum in the fourth quadrant and marked as $(1, -a)$ oe • x-axis intercepts marked as $(1+\frac{\sqrt{a}}{2}, 0)$ oe on the positive x-axis **and** $(1-\frac{\sqrt{a}}{2}, 0)$ oe on the negative x-axis • y-axis intercept marked as $(0, 4-a)$ oe Note: The 0's can be ignored (as shown in the diagram)
				Total 4 marks

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Question 17 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2024 · Question 23*

23	$\left(\frac{dy}{dx}=\right)2x-8$ or $(x-4)^2$….. or $(x=)\frac{-1+9}{2}$		5	M1 for using differentiation **or** completing the square **or** by symmetry
	$x=4$ or (4, –25) or (4, …) or $\frac{p+q}{2}=4$ oe			M1
	$q=-3$ **or** $p=11$			A1
	$(k=)(-3)^2-8(-3)-9$ **or** $(k=)(11)^2-8(11)-9$			M1
	Working required	24		A1 dep on M2
				Total 5 marks

23	$p^2-8p-9=k$ **and** $q^2-8q-9=k$ or $(p-9)(p+1)=k$ **and** $(q-9)(q+1)=k$		5	M1
ALT 1	$(q+14)^2-8(q+14)-9=q^2-8q-9$ **or** $(p-14)^2-8(p-14)-9=p^2-8p-9$ **or** $(q+14-9)(q+14+1)=(q-9)(q+1)$ **or** $(p-9)(p+1)=(p-14-9)(p-14+1)$ oe			M1 for a correct equation in one variable
	$q=-3$ **or** $p=11$			A1
	$(k=)(-3)^2-8(-3)-9$ **or** $(k=)(11)^2-8(11)-9$			M1
	Working required	24		A1 dep on M2
				Total 5 marks

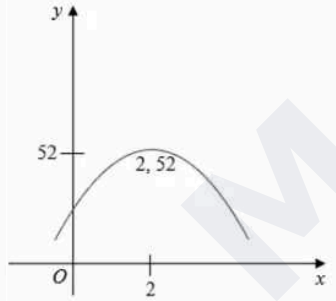
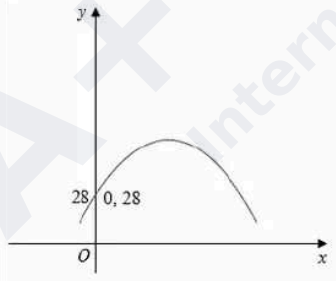
23	$(q+14)^2-8(q+14)-9(=k)$ **or** $(p-14)^2-8(p-14)-9(=k)$		5	M1
ALT 2	$(q+14)^2-8(q+14)-9=q^2-8q-9$ **or** $(p-14)^2-8(p-14)-9=p^2-8p-9$			M1 for a correct equation in one variable
	$q=-3$ **or** $p=11$			A1
	$(k=)(-3)^2-8(-3)-9$ **or** $(k=)(11)^2-8(11)-9$			M1
	Working required	24		A1 dep on M2
				Total 5 marks

23	$y=(x-4)^2-16-9$ oe or $k=(x-4)^2-25$ oe or $x^2-8x-9-k=0$		5	M1
ALT 3	$(x=)4\pm\sqrt{k+25}$ or $(x=)\frac{--8\pm\sqrt{(-8)^2-(4\times1\times(-9-k))}}{2\times1}$ oe or $(x=)\frac{8\pm\sqrt{100+4k}}{2}$ oe			M1
	$4+\sqrt{k+25}-(4-\sqrt{k+25})=14$ oe or $\frac{8+\sqrt{100+4k}}{2}-\left(\frac{8-\sqrt{100+4k}}{2}\right)=14$ oe			A1 for a correct equation in one variable
	$25+k=\left(\frac{14}{2}\right)^2$ oe			M1
	Working required	24		A1 dep on M2
				Total 5 marks

Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 25

Question	Working	Answer	Mark	Notes
25 (a)	$\pm 6\left(x \pm \frac{4}{2}\right)^2$ ..... or $\pm 6(x \pm 2)^2$ ..... **or** $\pm 6\left[(x \pm 2)^2 .....\right]$ or $\pm 6\left[\left(x \pm \frac{4}{2}\right)^2 .....\right]$ **or** $\pm 6\left(x \pm \frac{24}{2\times -6}\right)^2$		3	M1 for a start to completing the square **or** correct substitution into $a\left(x+\frac{b}{2a}\right)^2 + ...$ from the formula $a\left(x+\frac{b}{2a}\right)^2 - \frac{(b)^2}{4a} + c$
	$-6\left[\left(x-\frac{4}{2}\right)^2 - \left(\frac{4}{2}\right)^2\right]$ ..... or $-6\left[(x-2)^2 - 2^2\right]$ ..... **or** $-6\left[\left(x-\frac{4}{2}\right)^2 - \left(\frac{4}{2}\right)^2 .....\right]$ or $-6\left[(x-2)^2 - 2^2 .....\right]$ **or** $-6\left(x+\frac{24}{2\times -6}\right)^2 - \frac{24^2}{4\times -6}$			M1 for correctly completing the square but terms do not need to be simplified and 28 may or may not be present or correct simplification of the first two parts of $a\left(x+\frac{b}{2a}\right)^2 - \frac{(b)^2}{4a}(+c)$ **NB: Please refer to ALT mark scheme after (b) for comparison of coefficients method**
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$52-6(x-2)^2$		A1 oe eg $-6(x-2)^2+52$
25(a) **ALT**	$-bx^2 + 2bcx - bc^2 + a$ **and** $b = 6$ or $b = -6$		3	M1 for multiplying out $a - b(x-c)^2$ **and** $b = 6$ or $b = -6$
	$2bc = 24$ or $-bc^2 + a = 28$			M1 for equating coefficients
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$52-6(x-2)^2$		A1 oe eg $-6(x-2)^2+52$

(b)		'∩' or 'U' shaped symmetrical quadratic curve	3	B1 for drawing a '∩' or 'U' shaped symmetrical quadratic curve with the turning point in any quadrant
		Turning point marked as (2, 52)		B1 for drawing a '∩' shaped symmetrical quadratic curve in the correct quadrant with a turning point at (2, 52)
		Intersection with *y*-axis marked as (0, 28) or crossing at 28 marked		B1 for drawing a '∩' shaped symmetrical quadratic curve in the correct quadrant with an intersection on the *y*-axis marked as (0, 28) or marked as 28 on the *y*-axis
				Total 6 marks